Cloud Concepts and Connectivity — Guided Notes ANSWER KEY

Unit 1.7.2 | CompTIA Network+ N10-009 Obj. 1.7 | N10-008 1.7

For instructor use / distribute after students complete the notes.

Question 1

Match each cloud service model to what the customer is responsible for managing:

IaaS → _______

PaaS → _______

SaaS → _______

Answer: IaaS → OS, applications, data, access controls (everything from OS up); PaaS → application code and data; SaaS → access management (accounts, permissions, MFA) and data classification.

Question 2

Give one real-world example of each service model:

IaaS → _______

PaaS → _______

SaaS → _______

Answer: IaaS → AWS EC2, Azure Virtual Machines, Google Compute Engine; PaaS → Azure App Service, AWS Elastic Beanstalk, Google App Engine; SaaS → Microsoft 365, Google Workspace, Salesforce, Zoom (any valid example).

Question 3

A _______ cloud is shared infrastructure available to any paying customer. A _______ cloud is dedicated to a single organization. A _______ cloud combines both. A _______ cloud is shared among organizations with common requirements (e.g., government agencies).

Answer: Public; private; hybrid; community.

Question 4

A dedicated cloud connection (AWS calls it _______, Azure calls it _______) differs from a VPN connection because it _______. The tradeoff is _______.

Answer: Direct Connect; ExpressRoute; bypasses the public internet entirely, providing consistent bandwidth and predictable latency; significantly higher cost.

Question 5

In cloud networking, a _______ (VPC/VNet) is an isolated network environment. Within it, _______ divide the address space by purpose. _______ act as virtual firewalls attached to resources, controlling inbound and outbound traffic with stateful rules.

Answer: Virtual network; subnets; security groups.

Question 6

The shared responsibility model means that in IaaS, if an attacker exploits an unpatched OS vulnerability on a cloud VM, the responsibility belongs to _______. The cloud provider is responsible for _______.

Answer: The customer (they manage the OS and patching); the physical infrastructure, hypervisor, and network fabric.

Question 7

A cloud load balancer sits in front of multiple backend instances and provides: (1) _______, (2) _______, and (3) _______. It keeps the application available when individual instances _______.

Answer: Traffic distribution across healthy instances; health checks (routes away from failed instances); SSL/TLS termination; fail.

Question 8 — Real World Application

REAL WORLD: A teacher reports that Microsoft 365 is slow during first period but fast after 9 AM. The internet connection looks fine from the server room. What cloud-specific factors would you investigate?

Answer: 1. Check bandwidth utilization on the internet uplink during first period — 30+ students authenticating and syncing OneDrive simultaneously can saturate an undersized connection. 2. Check QoS policies — is Microsoft 365 traffic prioritized over general browsing? 3. Review DNS resolution time for M365 endpoints — slow DNS adds latency at connection establishment. 4. Check if the district uses a web proxy or content filter that may be bottlenecking M365 traffic. 5. Consider split tunneling if traffic routes through a VPN. Microsoft publishes recommended network configurations for M365 that include direct internet breakout for their endpoints.